

Alien ALN-9654 G Inlay

Ultra-high-performance general-purpose UHF RFID inlay powered by Alien Higgs-3 IC, optimized for high-dielectric materials such as windshields, reusable plastic pallets and containers. World Tag compliant for worldwide operation.



RFID SPECIFICATION

Protocol	ISO/IEC 18000-6C, EPCglobal Class 1 Gen 2
Integrated Circuit	Alien Higgs-3
Operating Frequency	840–960 MHz
EPC Size	96–480 Bits
User Memory	512 Bits
TID	32 Bits
Unique TID	64 Bits
Access Password	32 Bits
Kill Password	32 Bits
EPCglobal Certificate	950110126000001084

DRY INLAY DIMENSIONS

Antenna Width	3.66" [93.0 mm]
Antenna Length	0.748" [19.0 mm]
Web Width (~FR Unslit)	4.173" [106.0 mm]
Web Width (~FSR Slit)	3.799" [96.5 mm]
Web Pitch	1.0" [25.4 mm]
Inlays per Roll	12,500 Nominal
Maximum Roll OD	< 12" [304.8 mm]
Core ID	6" [152.4 mm]
Core Material	Fiberboard
Interleaf Material	Paper
Interleaf Width	1.5" [38.1 mm]

WET INLAY DIMENSIONS

Inlay Width	3.799" [96.5 mm]
Inlay Length	0.913" [23.2 mm]
Web Width	3.925" [99.7 mm]
Web Pitch	1.0" [25.4 mm]
Inlays per Roll	12,500 Nominal
Maximum Roll OD	< 16" [406.4 mm]
Printable Surface	White TT Printable White Film
Overlay Adhesive	General Purpose Permanent
Inlay Adhesive	General Purpose Permanent
Adhesive Application Temp	> +25°F [-4°C]
Adhesive Service Temp	-40°F to +200°F [-40°C to +93.3°C]
Release Liner	40# SCK

ENVIRONMENTAL

Shelf Life (Dry)	5 years at +25°C @ 40% RH
Shelf Life (Wet)	2 years at +25°C @ 40% RH
Recommended Storage	+25°C @ 40% RH
Storage Limits	-25°C to +50°C, 20–90% RH non-condensing
Operating Limits	-40°C to +70°C, 20–90% RH non-condensing
Bend Diameter	> 1.97" [50 mm]
Write Cycles	100,000 @ 25°C
RoHS	2002/95/EC, 2005/618/EC, 2011/65/EU Compliant
REACH	1907/2006/EC Compliant
ESD Limit (HBM / CDM)	5.0 kV / 1.5 kV

FEATURE HIGHLIGHTS

- Alien Higgs-3 IC with 800-bit nonvolatile memory
- 96–480-bit extensible EPC memory bank
- 512-bit User Memory for distributed data applications
- Pre-programmed 64-bit Unique TID for authentication and serialization
- Password-protected read/write with 32-bit Access and Kill passwords
- World Tag compliant – worldwide UHF operation 840–960 MHz
- Available in dry (unslit/slit) and white wet inlay roll formats

PRODUCT GROUP

Category	UHF RFID Inlay
Product Group	RFID Label / Inlay